

Joint Optimization of Power Allocation and Load Balancing for Hybrid VLC/RF Networks

Mohanad Obeed,  Anas M. Salhab,  Salam A. Zummo, and Mohamed-Slim Alouini 

Abstract—In this paper, we propose and study a new joint load balancing (LB) and power allocation (PA) scheme for a hybrid visible light communication (VLC) and radio frequency (RF) system consisting of one RF access point (AP) and multiple VLC APs. An iterative algorithm is proposed to distribute users on APs and distribute the powers of the APs on their users. In the PA subproblem, an optimization problem is formulated to allocate the power of each AP to the connected users for total achievable data rate maximization. In this subproblem, we propose a new efficient algorithm that finds optimal dual variables after formulating them in terms of each other. This new algorithm provides faster convergence and better performance than the traditional subgradient method. In addition, it does not depend on the step size or the initial values of the variables, which we look for, as the subgradient does. Then, we start with the user of the minimum data rate seeking another AP that offers a higher data rate for that user. Users with lower data rates continue reconnecting from one AP to another to balance the load only if this travel increases the summation of the achievable data rates and enhances the system fairness. Two approaches are proposed to have the joint PA and LB performed: a main approach that considers the exact interference information for all users, and a suboptimal approach that aims to decrease the complexity of the first approach by considering only the approximate interference information of users. The numerical results demonstrate that the proposed algorithms improve the system capacity and system fairness with fast convergence.

Index Terms—Hybrid visible light communication and radio frequency (VLC/RF); Load balancing; Power allocation.

I. INTRODUCTION

The growing demand for high data rate and license-free spectrum applications has recently stimulated many researchers to investigate visible light communication (VLC) as a promising technique for indoor communication. As is known, radio frequency (RF) communications are becoming more restricted because of the limited spectrum resources in wireless networks. As a solution, VLC has

been introduced in indoor environments to overcome RF limitations and provide better services to users.

VLC is a communication system that uses light emitting diodes (LEDs), which are highly energy efficient, as transmitters to emit both light and information signals to users. The receiver must be equipped with a photodetector device that converts the light signal to an electrical signal. Hence, data are transmitted using an intensity modulation (IM) scheme at the transmitter and received using a direct detection (DD) scheme at the receiver (IM/DD). In addition, for being license-free and secure systems, VLCs are known for their ability to reject RF interference and their high bandwidth potential [1]. On the other hand, VLC suffers from blockages. Unlike the RF network, VLC works properly only when the line of sight (LoS) component between the transmitter and the receiver is available. Therefore, supplementing the VLC network with an RF access point (AP) increases the network coverage and improves the total system capacity [2].

Authors of Ref. [3] considered the cooperation between two VLC APs to increase the rate region by controlling the transmission power. In Ref. [4], the authors studied the advantages of combining RF and VLC APs by showing how this combination enhances the system throughput and decreases the delay. In Ref. [5], the authors showed that the users are distributed dynamically on both VLC and RF networks based on the user channel condition. Then, users with lower achievable rates are migrated to the RF system to increase their data rates. To decrease the number of handovers, [6] proposed a dynamic load balancing (LB) scheme where the quasi-static users are connected to VLC APs, while the unfixed users are connected to RF APs. To study the joint load balancing and time resource allocation, [7] proposed a bidirectional allocation game where each user is initially serviced from the AP that provides the highest signal-to-noise ratio (SNR) with time resource allocation. Unsatisfied users are then reconnected to an adjacent AP only in the case that it can offer a better service. The procedures of resource allocation and load balancing are repeated until no more handover occurs. In Ref. [8], power and bandwidth allocation was investigated for an energy-efficient hybrid VLC/RF system consisting of only one VLC AP and one RF AP. Since all the users (either served by RF AP or VLC AP) do not suffer from interference, the problem formulated in Ref. [8] is easy to tackle. Authors of Ref. [9] studied how the APs should cooperate to mitigate interference with balancing the load, but the power allocation problem

Manuscript received August 10, 2017; revised February 5, 2018; accepted February 15, 2018; published April 19, 2018 (Doc. ID 304099).

Mohanad Obeed, A. M. Salhab (e-mail: salhab@kfupm.edu.sa), and S. A. Zummo are with the Department of Electrical Engineering, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia.

M.-S. Alouini is with the Computer, Electrical, and Mathematical Sciences and Engineering Division, King Abdullah University of Science and Technology, Thuwal 23955-6900, Saudi Arabia.

<https://doi.org/10.1364/JOCN.10.000553>

has not been considered. The authors of Refs. [10] and [11] extended the work of Ref. [9] by studying the cell formation with allocating the power for only a VLC network. However, problems of associating users and allocating power have been tackled separately in Refs. [10] and [11], where they are interlinked problems and must be tackled jointly. To the best of the authors' knowledge, joint LB and power allocation (PA) for a hybrid VLC/RF network have not been presented or studied before.

In this paper, different from the above literature mentioned, we study the two problems of PA and LB in a hybrid VLC/RF network for the sake of data rate maximization and system fairness improvement. The network consists of multiple VLC APs and one RF AP. First, each user is connected to its closest AP. Then, each AP performs its optimization problem (allocates the power for the associated users) in order to maximize the summation of the achievable data rates per AP. After that, users with lower data rates start reconnecting from the AP to the other to balance the load only in the case that this transfer increases the summation of the achievable data rates. This transfer of users continues until no improvement in the summation of data rates is achieved. We prove the convergence of the proposed algorithm analytically and numerically. The inter-cell interference makes the joint PA and LB problem very difficult. Therefore, two approaches are proposed to have the joint PA and LB implemented: 1) the approach that considers instantaneous (exact) interference information for all users, 2) and a suboptimal approach that aims to decrease the complexity of the first approach by considering only the approximate interference information of users. This work is different from the work of Ref. [7] in several aspects, as follows: 1) we generalize the system to include an RF AP, 2) in the PA subproblem, the authors in Ref. [7] performed a time resource allocation problem, while we allocate the power for the users, 3) we formulate the PA problem as a mathematical optimization problem and not by using fuzzy logic as in Ref. [7], and 4) we propose a suboptimal approach to decrease the complexity of the procedures significantly with a negligible loss in the performance. The procedures are simplified in such a way when a user connection is transferred; only two APs perform the power allocation problem and not all the APs as in Ref. [7]. In the PA optimization subproblem, in VLC and RF APs, we formulate the problem for maximizing the summation of achievable users' data rates under certain quality-of-service (QoS) constraints. These QoS constraints are formulated to control the trade-off between the system capacity and the system fairness. For given interference information, the PA problem is proved to be concave but is not easy to tackle. Similar PA problems were studied before (see [8,12–16]). In finding dual variables, the authors of these references used the subgradient method, which is very sensitive to step size selection and needs a large number of iterations for convergence. Here, we derive a new efficient algorithm that finds the optimal dual variables after formulating them in terms of each other without requiring optimization of the step size or selecting the initial values carefully. This new algorithm provides faster convergence and better performance than the traditional subgradient method.

The rest of this paper is organized as follows. The system and channel models are introduced in Section II. In Section III, we present the problem formulation and proposed algorithms. Some simulation results are presented and discussed in Section IV. Finally, the paper is concluded in Section V.

II. SYSTEM AND CHANNEL MODELS

The system under consideration consists of N_{ap} VLC APs, one RF AP, and N_u users, as shown in Fig. 1. The users are distributed uniformly in the room, and the APs are fixed in the ceiling of the room. Each VLC AP is equipped with multiple LEDs that use IM to transmit the light signal to the users, which receive the light by a photodetector (PD). The RF AP is assumed to cover all of the room area. Also, user locations are assumed to be unchanged during a short period of interest T . Thus, the VLC and RF channel-state information of both VLC and RF links is considered to be constant during this period. We assume that the maximum available bandwidth at the AP i is divided fairly among all users connected to that AP.

A. VLC Channel

The LoS VLC channel between the i th LED and the j th user can be modeled as follows [17]:

$$h_{j,i}^{(v)} = \frac{(m+1)A_p}{2\pi d_{j,i}^2} \cos^m(\phi) g_{of}(\theta) \cos(\theta), \quad (1)$$

where m is the Lambertian index that is given by $m = -1/\log_2(\cos(\theta_{1/2}))$, where $\theta_{1/2}$ is the half intensity radiation angle, A_p is the physical area of the receiver PD, $d_{j,i}$ is the distance between the i th AP to the j th user, g_{of} is the gain of the optical filter, ϕ is the angle of radiance at the AP, θ is the angle of incidence at the PD, and $f(\theta)$ is the optical concentrator gain, which is a function of θ that is given by

$$f(\theta) = \begin{cases} \frac{n^2}{\sin^2(\Theta)}, & 0 \leq \theta \leq \Theta, \\ 0, & \theta > \Theta, \end{cases} \quad (2)$$

where n is the refractive index, and Θ is the semi-angle of the field of view (FoV) of the PD. In a VLC network, the

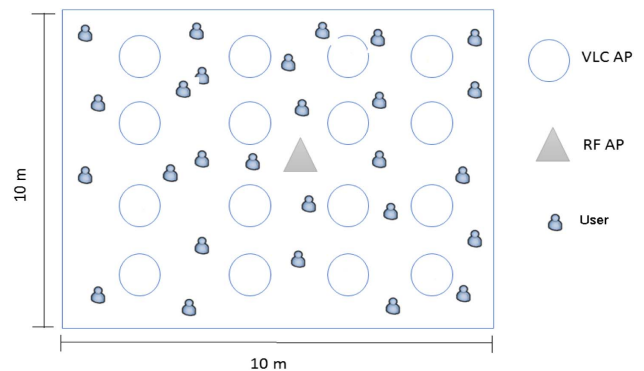


Fig. 1. System model.

LED has to operate in the linear region so that the optical power at its output is a linear function of the input voltage. In VLC networks, the signal deteriorates significantly in non-LoS VLC transmissions that might lead to unsuccessful data transmission [7–9], so we work only on LoS paths. This assumption does not affect the proposed algorithm since it does not depend on a specific channel model. In other words, whatever the channel model is, the proposed algorithm for the power allocation will give us the optimal solution, as will be shown. Therefore, when the LoS path is available, $h_{j,i}^{(v)}$ is given by Eq. (1); otherwise, $h_{j,i}^{(v)} = 0$. The probability of the availability of the LoS from an AP to a user is denoted by α , and it is assumed to be uniformly distributed. It was shown in Ref. [8] that the average electrical power of the received signal at user j from the VLC AP i is calculated as follows:

$$P_{\text{elec},j,i} = \left(\rho \sqrt{P_{j,i}^{(v)}} h_{j,i}^{(v)} \right)^2, \quad (3)$$

where ρ is the optical-to-electric conversion efficiency, and $P_{j,i}^{(v)}$ is the allocated power for user j from the AP i , where $P_{i,\max}^{(v)}$ is the maximum transmitted optical power of the AP i . The received SNR at user j from the VLC AP i can be expressed as

$$\Gamma_{j,i}^{(v)} = \frac{\left(\rho \sqrt{P_{j,i}^{(v)}} h_{j,i}^{(v)} \right)^2}{(B_{i,\max}^{(v)}/N_i)N_0^{(v)} + \rho^2 \sum_{l=1, l \neq i}^{N_p} P_{B_{j,l}}^{(v)} h_{j,l}^{(v)2}}, \quad (4)$$

where $B_{i,\max}^{(v)}$ is the maximum available bandwidth at the AP i , N_i is the number of users connected to the AP i , $N_0^{(v)}$ is the noise power spectral density, and $P_{B_{j,l}}^{(v)}$ is the interference power at user j caused by the AP l . It is worth noting that $P_{B_{j,l}}^{(v)}$ must be calculated carefully since it represents the power allocated from the AP l to the same frequency spectrum that is allocated for user j . For instance, if AP l is associated with four users and user j along with another user are connected to AP i , $P_{B_{j,l}}^{(v)}$ should be calculated as the summation of the powers of two users out of the four users that are connected to the AP l and use the same spectrum frequency used by user j .

Here, Shannon's capacity equation is used to express the achievable data rate that represents a limit for the performance without considering specific modulation and coding schemes. This equation is employed due to its mathematical tractability, since using different modulation schemes does not significantly affect the optimality of the achieved data rate [18]. Therefore, according to Refs. [8] and [9], the maximum data rate that can be achieved at user j from the VLC AP i is given by

$$R_{j,i}^{(v)} = (B_{i,\max}^{(v)}/N_i) \log_2(1 + \Gamma_{j,i}^{(v)}). \quad (5)$$

B. RF Channel

As mentioned before, the RF cell is assumed to cover all the room area and a spectrum of a total bandwidth $B_{\max}^{(r)}$ at

the RF AP is considered to be flat fading and can be partitioned into nonoverlapping channels of unequal bandwidths. According to Ref. [2], the RF channel gain between user j and the RF AP is expressed as

$$h_j^{(r)} = \sqrt{10^{-\frac{L(d)}{10}}} \left(\sqrt{\frac{K}{K+1}} h_d + \sqrt{\frac{1}{K+1}} h_s \right), \quad (6)$$

where $K = 10$ dB is the Rician factor for indoor 60 GHz mm Wave links, $h_d = \sqrt{0.5}(1+j)$ is the LoS fading channel, h_s is a circular normal distributed with zero mean and unity variance and it is the fading channel of the scattered path, and $L(d)$ is the corresponding large-scale fading loss in decibels at the separation distance d , given by

$$L(d) = L(d_0) + 10v \log_{10}(d/d_0) + Z, \quad (7)$$

where $L(d_0) = 68$ dB is the reference path loss at some reference distance $d_0 = 1$ m, $v = 1.6$ is the path loss exponent, and Z is the shadowing component, which is assumed to be a zero-mean Gaussian-distributed random variable with a standard deviation of 1.8 dB. The shadowing effect induced by human bodies in the proximity of the mmWave radio links is omitted [2]. The data rate achieved by the RF link between user j and the RF AP can be expressed as

$$\Gamma_j^{(r)} = \frac{P_j^{(r)} |h_j^{(r)}|^2}{(B_{\max}^{(r)}/N_r)N_0^{(r)}}, \quad (8)$$

where $B_{\max}^{(r)}$ is the maximum available bandwidth at the RF AP, and N_r is the number of users connected to the RF AP. The achievable data rate that can be achieved by user j from the RF AP is given by

$$R_j^{(r)} = (B_{\max}^{(r)}/N_r) \log_2(1 + \Gamma_j^{(r)}). \quad (9)$$

III. PROBLEM FORMULATION

The problem here is how to assign all the users to the APs and how to allocate the AP powers for the assigned users aiming to maximize the system capacity with keeping the fairness between the users at an acceptable level. Therefore, we start by assigning the users based on their distance from the APs, where each user selects the closest AP to it. Then, each AP performs its own power allocation problem for its associated users.

The achievable data rate of user j is expressed as

$$R_{j,i} = \begin{cases} R_{j,i}^{(v)}, & i \in \mathcal{C}_v, \\ R_{j,i}^{(r)}, & i \in \mathcal{C}_r, \end{cases} \quad (10)$$

where $\mathcal{C}_v$ denotes the set of the VLC APs, and $\mathcal{C}_r$ denotes the RF AP.

A. Power Allocation in VLC and RF Access Points

In this section, we aim to allocate the power for the N_i users that are connected to the AP i . The objective function here is to maximize the summation of the users' data rates under certain QoS constraints. These constraints are

formulated to guarantee some fairness for users per AP. We formulate the optimization problem as a general form for the VLC APs or the RF APs. Hence, $[P_{j,i} \ B_{j,i} \ h_{j,i} \ N_0] = [P_{j,i}^{(v)} \ (B_{i,\max}/N_i) \ \rho h_{j,i}^{(v)} \ N_0^{(v)}]$, if the AP i is a VLC AP, and if the AP i is an RF AP, so $[P_{j,i} \ B_{j,i} \ h_{j,i} \ N_0] = [P_{j,i}^{(r)} \ (B_{\max}/N_r) \ h_{j,i}^{(r)} \ N_0^{(r)}]$. Therefore, the optimization problem of the AP i can be expressed as

$$\max_{P_{1,i}, \dots, P_{N_i,i}} \sum_{j=1}^{N_i} R_{j,i}, \quad (11a)$$

$$\text{s.t. } R_{j,i} \geq \gamma_{j,i}, \quad j = 1, \dots, N_i, \quad (11b)$$

$$\sum_{j=1}^{N_i} P_{j,i} \leq P_{i,\max}, \quad (11c)$$

$$P_{j,i} \geq 0 \quad \forall j, \quad (11d)$$

where N_i is the number of users associated with the AP i , and $\gamma_{j,i}$ is the minimum data rate that can be achieved at user j from the AP i , which is given by

$$\gamma_{j,i} = \beta \frac{B_{i,\max}}{N_i} \log \left(1 + \frac{(P_{i,\max}/N_i) |h_{j,i}|^2}{(B_{i,\max}/N_i) N_0 + X_{j,i}} \right), \quad (12)$$

where $X_{j,i}$ is the interference term of user j and is equal to $\rho^2 \sum_{l=1, l \neq i}^{N_p} P_{B_{j,l}}^{(v)} h_{j,l}^{(v)2}$ if the AP i is VLC, and in the case of RF AP, $X_{j,i} = 0$, and β is a value in the interval $[0,1]$. If we select $\beta = 1$, this means that the AP will distribute its resources fairly (regardless of user channels), where each user gains the same power $P_{i,\max}/N_i$ and same bandwidth $B_{i,\max}/N_i$. On the other hand, if β is close to zero, this means that the objective function is released from the first constraint, which leads to an increase in the AP capacity.

Jain's fairness index is used to measure the AP fairness and the whole system fairness. Therefore, the fairness of the AP i is given by

$$F_i = \frac{\left(\sum_{j=1}^{N_i} R_{j,i} \right)^2}{N_i \sum_{j=1}^{N_i} R_{j,i}^2}, \quad (13)$$

and the fairness of the system is given by

$$F_s = \frac{\left(\sum_{i=1}^{N_{ap}} \sum_{j=1}^{N_i} R_{j,i} \right)^2}{N_{ap} \sum_{i=1}^{N_{ap}} \sum_{j=1}^{N_i} R_{j,i}^2}. \quad (14)$$

The problem in Eq. (11) is not easy to tackle since in the case where the AP i is a VLC AP, the interference term in the objective function makes the problem difficult. On the other hand, if the interference terms are given, the problem in Eq. (11) becomes a concave problem and can be solved by using the Lagrangian dual problem. First, we solve this problem under the assumption that the interference terms are given, then we provide an iterative algorithm that achieves the optimal power allocation. Obviously, the constraints in Eqs. (11c) and (11d) are linear functions, while the objective function and the constraint in Eq. (11b) are well known as concave functions, as shown

in Refs. [8] and [13]. Therefore, the optimization problem in Eq. (11) is a concave problem with one global optimum solution. Hence, we can use the dual problem to achieve the optimal solution, where the strong duality holds in the concave problems. The dual optimization problem of the problem in Eq. (11) can be expressed as follows:

$$\begin{aligned} \zeta = & - \sum_{j=1}^{N_i} R_{j,i} - \sum_{j=1}^{N_i} \mu_{j,i} (R_{j,i} - \gamma_{j,i}) \\ & + v_i \left(\sum_{j=1}^{N_i} P_{j,i} - P_{i,\max} \right), \end{aligned} \quad (15)$$

where μ_j is the Lagrangian multiplier for the data rate constraint of the j th user, and v_i is the Lagrangian multiplier for the total power constraint.

In the following, an efficient algorithm is proposed to solve the problem in Eq. (11) by minimizing the dual problem in Eq. (15). From the Karush–Kuhn–Tucker (KKT) conditions [19], we have

$$\frac{\partial \zeta}{\partial P_{j,i}} = 0, \quad (16)$$

where

$$P_{j,i} = \frac{(1 + \mu_{j,i}) B_{j,i}}{v_i \ln(2)} - \frac{N_0 B_{j,i} + X_{j,i}}{|h_{j,i}|^2}, \quad (17)$$

where the variable $\mu_{j,i}$ must guarantee that the constraint in Eq. (11b) is feasible, and v_i must guarantee that the constraint in Eq. (11c) is feasible too. Now, these dual variables must be found to obtain $P_{j,i} \ \forall j$. In several papers, such as [8,12–16], the authors found such dual variables by using the gradient decent method, which is an iterative algorithm that needs a large number of iterations to converge, a very careful selection of the step size, and a careful initial value selection for the dual variables. Proposing another approach that gets rid of these requirements (optimizing step size and the careful selection of the initial values) significantly simplifies the problem and provides a better performance. Here, we find a closed-form expression for v_i in terms of $\mu_{j,i}$ and vice versa. Then, we solve them alternatively until they converge. First, it is trivial showing that the constraint in Eq. (11c) must hold with equality at optimality. Otherwise, we can increase one of the power variables until the constraints hold with equality, which leads to increasing the objective function, and hence, contradicting the optimality. By substituting Eq. (17) in Eq. (11c), we have

$$v_i = \frac{\sum_{j=1}^{N_i} B_{j,i} (1 + \mu_{j,i})}{\ln(2) \left(P_{i,\max} + \sum_{j=1}^{N_i} \frac{B_{j,i} N_0 + X_{j,i}}{|h_{j,i}|^2} \right)}. \quad (18)$$

Similarly, by substituting Eq. (17) in Eq. (11b), $\mu_{j,i}$ must be

$$\mu_{j,i} \geq \frac{v_i \ln(2) P_{j,i}^{\min}}{B_{j,i}} + \frac{v_i \ln(2) (B_{j,i} N_0 + X_{j,i})}{B_{j,i} |h_{j,i}|^2} - 1, \quad (19)$$

where $P_{j,i}^{\min}$ is given by

$$P_{j,i}^{\min} = \frac{(2^{\gamma_{j,i}/B_{j,i}} - 1) (B_{j,i} N_0 + X_{j,i})}{|h_{j,i}|^2}, \quad j = 1, \dots, N_i, \quad (20)$$

which is the minimum required power to achieve the constraint in Eq. (11b). Hence, Eqs. (18) and (19) depend on each other and can be solved alternatively starting from an initial value of one of them until they converge. After that, Eq. (17) is used to find the optimal power allocation. Algorithm 1 is proposed to solve the optimization problem in Eq. (11) with given interference information. Condition 5 in Algorithm 1 examines the convergence of all dual variables v_i and $\mu_{j,i}$, $j = 1, 2, \dots, N_i$.

Algorithm 1: Power allocation for the AP i

1. Input $B_{j,i}$, $\mu_{j,i}(0) \forall j$.
 2. for $q = 1:M$
 3. Find $v_i(q)$ from Eq. (18) and $\forall j$, calculate $P_{j,i}$.
 4. For all j , check if the calculated $P_{j,i} \geq P_{j,i}^{\min}$. If so, $\mu_{j,i}(q) = \mu_{j,i}(q-1)$; otherwise, calculate $\mu_{j,i}(q)$ from Eq. (19) by equating both sides, and then update $P_{j,i}$.
 5. If $|v_i(q) - v_i(q-1)| \leq \epsilon$, break;
 6. end for
 7. Find $P_{j,i} \forall j$ using Eq. (17).
-

1) *Convergence Analysis:* Here, we analyze Algorithm 1 in terms of convergence. Before we start analyzing the convergence, we should note that the values of $P_{j,i} \forall j$ after Step 3 are different from the values of $P_{j,i} \forall j$ after Step 4 (i.e., the values of $P_{j,i} \forall j$ change twice in the same iteration). Specifically, in Step 3, the power values are changed because of updating the value of v_i , while in Step 4, the power values are changed because of updating the values of the $\mu_{j,i}$ s. To avoid this confusion, we denote the values of $P_{j,i} \forall j$ after Step 3 by $P_{j,i}^{(3)}$ and denote the values of $P_{j,i} \forall j$ after Step 4 by $P_{j,i}^{(4)}$.

At any q th iteration, it can be shown that in Step 3, the variable $v_i(q)$ steers the summation of powers to be equal to $P_{i,\max}$ [i.e., to achieve the constraint in Eq. (11c) with equality]. In other words, if we find $v_i(q)$ at Step 3 and substitute it in Eq. (17) $\forall j$, we find that $\sum_{j=1}^{N_i} P_{j,i}^{(3)}(q) = P_{i,\max}$. On the other hand, Step 4 implies that each $P_{j,i}^{(3)}$ that is less than the corresponding $P_{j,i}^{\min}$ [i.e., each $P_{j,i}^{(3)}$ that does not satisfy the corresponding constraint in Eq. (11b)] is increased by increasing the associated $\mu_{j,i}$ to achieve that $P_{j,i}^{(4)}(q) = P_{j,i}^{\min}$. This increase in power yields a violation of the constraint in Eq. (11c) as $\sum_{j=1}^{N_i} P_{j,i}^{(4)}(q) \geq P_{i,\max}$. Hence, in the $(q+1)$ th iteration, $v_i(q+1)$ will be greater than $v_i(q)$ to have that $\sum_{j=1}^{N_i} P_{j,i}^{(3)}(q+1) = P_{i,\max}$ again.

Without loss of generality, we assume that in the q th iteration, $P_{j,i}(q) < P_{j,i}^{\min}$ for $j = 1, \dots, k-1$ and $P_{j,i}(q) \geq P_{j,i}^{\min}$ for $j = k, \dots, N_i$. From Eq. (17), we note that the increase in v_i leads to decreasing each $P_{j,i} \forall j$ while keeping the constraint $\sum_{j=1}^{N_i} P_{j,i}^{(3)}(q+1) = P_{i,\max}$ satisfied. Hence, we have that $P_{j,i}^{(3)}(q+1) < P_{j,i}^{(3)}(q)$, $j = k, \dots, N_i$ (because Step 4 causes no change in these powers in the iteration q), and $P_{j,i}^{(3)}(q+1) > P_{j,i}^{(3)}(q)$, $j = 1, \dots, k-1$ to keep the constraint $\sum_{j=1}^{N_i} P_{j,i}^{(3)}(q+1) = P_{i,\max}$ satisfied. This means that $\mu_{j,i}(q+1)$ must be increased to achieve the constraint

$P_{j,i}^{(4)}(q+1) = P_{j,i}^{\min}$, $j = 1, \dots, k-1$ but with a lower amount than what was required in $\mu_{j,i}(q)$. Consequently, with implementing one iteration more, we have

$$\mu_{j,i}(q+2) - \mu_{j,i}(q+1) < \mu_{j,i}(q+1) - \mu_{j,i}(q), \quad (21)$$

$$j = 1, \dots, k-1.$$

Similarly,

$$v_i(q+2) - v_i(q+1) < v_i(q+1) - v_i(q). \quad (22)$$

Therefore, as the number of iterations increases, the amount of change in v_i and all $\mu_{j,i} \forall j$ approaches zero. Thus, Algorithm 1 is convergent.

It is also important to note that in Eq. (19), v_i is a factor of the expression $1 + \mu_{j,i}$, which means that starting with any initial values for $\mu_{j,i}$ will be compensated by v_i to have the same value of $P_{j,i}$ as Eq. (17) shows. Hence, Algorithm 1 does not depend on the initial values.

2) *Finding the Exact Interference Information:* Now, in the VLC APs, the problem is how to find the instantaneous interference ($X_{j,i}$) of each user, which is difficult to find because the term $X_{j,i}$ of user j depends on all powers of the APs that are allocated for the j th user frequency spectrum. Therefore, we provide an iterative algorithm that solves the PA problem of all the VLC APs with finding the instantaneous interference of each user.

Algorithm 2 provides the optimal power allocation for a given distributed user. It is worth stating that the RF AP performs its power allocation using Algorithm 1 and it is not included in Algorithm 2. This is because the interference of the users connected to the RF AP is zero. In Algorithm 2, the only step that needs to exchange the information between APs is Step 3 (calculating the interference). To calculate the interference $X(j, i)$, the AP i must know the power allocated from the other APs for the BW used by user j . Therefore, the APs must exchange their power information to have the interference information at each user.

Algorithm 2: Power allocation for all of the VLC APs with a given number of distributed users

1. Each AP allocates the power for the users equally.
 2. for $q = 1:M$
 3. Calculate $X_{j,i}^{(q)}$ for all users in the system.
 4. Perform Algorithm 1 for all APs.
 5. if $\sum_{i=1}^{N_p} \sum_{j=1}^{N_i} (X_{j,i}^{(q)} - X_{j,i}^{(q-1)})^2 \leq \epsilon$; break;
 6. End for
-

B. Load Balancing

As stated earlier, each user is initially connected to its closest AP. Then, each AP performs its own power allocation problem, as shown in the previous section. However, some APs will be overloaded, which may cause some users connected to these APs to receive a poor QoS. Therefore, after the PA optimization problem (Algorithm 2) is performed, the user with the poorest QoS is reconnected

to either the RF AP or another adjacent VLC AP if the later can provide a better service and increase the system capacity. The users connected to the RF AP do not receive any interference so that the strongest candidate AP for the user (which needs to reconnect to another AP) to travel to is the RF AP. The approach here is to arrange the users as a queue starting from the user with the lowest QoS up to the highest quality serviced user. Then, each user in its turn tests if the RF AP can provide a better achievable data rate for it or not. If so, the user migrates to the RF AP; otherwise, the user transfers to another adjacent AP if that AP can provide a better achievable data rate for him. These procedures continue until no improvement in the system capacity can be achieved. From the PA problem in Eq. (11), each VLC AP offers for each user its QoS denoted by $\gamma_{j,i}$ which is a function of the number of users connected to that AP, as shown in Eq. (12). Hence, the maximum offered achievable data rate for user j that is connected to the AP i is given by

$$\bar{R}_{j,k} = \max_{k \in \chi_i} (\gamma_{j,k}), \quad (23)$$

where χ_i is the set of the APs that are very close to the AP i , and $N_k < N_i - 1 \forall k$. Since $\gamma_{j,k}$ depends on the value of β , if $N_k = 0$, that means the AP k offers all its resources for the coming user, and hence, $\beta = 1$ in Eq. (23). Otherwise, if $N_k \geq 1$, β is determined by the AP k . It is important to state that the transfer of a user connection changes the interference information, which enforces the system to reimplement Algorithm 2 with each transfer. This process of transfer continues only if there is an improvement in the system capacity. To prove the convergence of this approach, first, we should note that the step of sorting users is conducted at the beginning of every round of testing all users, not with each user transfer. In addition, we cancel each user transfer from one AP to another that produces a degradation in the system capacity. Consequently, with each user transfer the system capacity increases, and as we know the capacity has a limit, which means that the convergence occurs when we approach that capacity limit.

C. Suboptimal Approach: Averaging the Interference

The disadvantage of the above approach is its high complexity since, with each user connection transfer, all the APs must perform the power allocation problem, which is highly complex. The reason behind that is the need for exact interference information to implement both the PA and the LB together. In this section, we aim to simplify the problem solution by relaxing this demand. User j experiences interference from each AP with a power that might be less or greater than the average power. This average power is calculated under the assumption that each AP distributes its power equally for its associated users. Therefore, the essence of the approach here is that instead of obtaining the instantaneous interference, we obtain the average interference that is calculated by averaging the power of the interference coming from all the APs. Therefore, the approximate inter-cell interference at user j that is connected to the AP i is given by

$$X_{j,i} = \sum_{l=1, l \neq i}^{N_p} \frac{P_{l,\max}}{N_i} h_{j,l}^2. \quad (24)$$

The above assumption significantly simplifies the problem as there is no need to implement the PA optimization at all the APs at each user connection transfer. In other words, we only need to perform the PA optimization problem (Algorithm 1) at only two APs (the departed from and the arrived at APs).

IV. SIMULATION RESULTS

In this section, we verify the capability of the proposed algorithms for enhancing the performance of the hybrid VLC/RF network. We show the convergence of the proposed algorithms and how they increase the system capacity. A 10×10 room area is assumed with 16 VLC APs and 1 RF AP fixed in the ceiling. The values of all parameters in the considered VLC/RF system are given in Table I. A Monte Carlo simulation is used to assess the performance of the proposed algorithms, where in each simulation iteration, a uniform random number is generated between 0 and 1. If that number is less than α (the probability of the availability of the LoS), the LoS component is available; otherwise, the LoS component is not available.

In Figs. 2 and 3, we show the performance and convergence of the proposed Algorithm 1 implemented at AP i and compare it with the subgradient method. Both figures should be analyzed together because Fig. 2 shows the maximum constraint violation of the approaches (i.e., how much the approaches are close from achieving the constraints versus the number of iterations), while Fig. 3 shows the value of the objective function for the different approaches versus the number of iterations. It is observed that the subgradient method with large step size, such as step size = 0.5, achieves the constraints slightly faster than the proposed approach, as shown in Fig. 2, but it cannot achieve the same system capacity as the proposed approach can achieve, as shown in Fig. 3. In other words, if we assume that the violation tolerance is 10^{-4} , both approaches (the proposed and the subgradient with large step size) almost satisfy the constraints after 100 iterations,

TABLE I
SIMULATION PARAMETERS

Name of the Parameter	Value of the Parameter
Room height	3 m
Maximum bandwidth of VLC AP, $B_{\max}^{(v)}$	30 MHz
Maximum bandwidth of RF AP, $B_{\max}^{(r)}$	30 MHz
Physical area of a PD, A_p	0.1 cm ²
Half-intensity radiation angle, $\theta_{1/2}$	60°
FoV semi-angle of the PD, Θ	90°
Gain of optical filter, g_{of}	1
Refractive index, n	1.5
Optical-to-electric conversion efficiency, ρ	1
Transmitted power of the VLC AP, $P_{\max}^{(v)}$	4 W
Transmitted power of the RF AP, $P_{\max}^{(r)}$	2 W
Noise power spectral density of LiFi, $N_0^{(v)}$	10^{-21} A ² /Hz
Variance of AWGN in RF AP, $N_0^{(r)}$	10^{-19} A ² /Hz

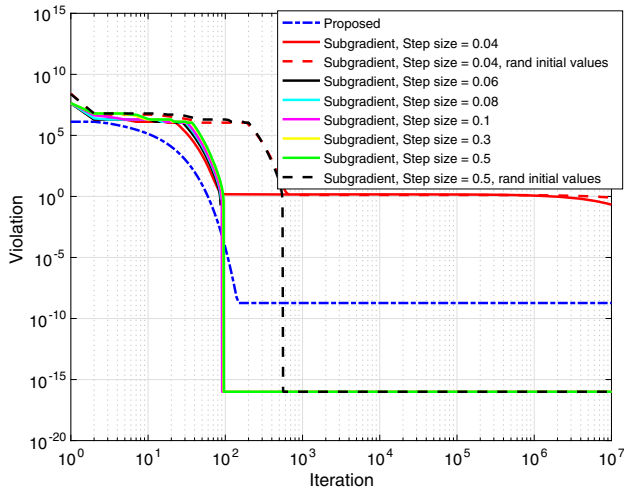


Fig. 2. Comparison between the proposed Algorithm 1 and the subgradient method with different step sizes by plotting the violation of the constraints versus the number of iterations for AP i , $P_{i,\max} = 10$ W.

while the proposed approach provides higher AP capacity than the subgradient method, as shown in Fig. 3. On the other hand, the subgradient method with small step size, such as step size = 0.04, starts to satisfy the constraints after more than 10^7 iterations, as shown in Fig. 2, and after that huge number of iterations, it starts approaching the capacity achieved by the proposed approach, as shown in Fig. 3. In addition, we implement the subgradient method with random initial values of the dual variables to show that the subgradient methods depend highly on the selected initial values, while the proposed algorithm does not. It is important to note that the disadvantage of the subgradient method is that the step size must be optimized to have the best performance, which is an additional problem that complicates the already subgradient method. Besides, the initial values for the dual variables must be

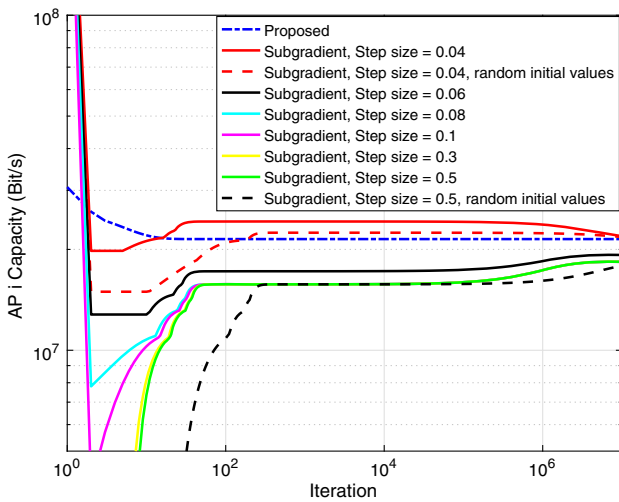


Fig. 3. Comparison between the proposed Algorithm 1 and the subgradient method with different step sizes by plotting the system capacity versus the number of iterations, $P_{i,\max} = 10$ W.

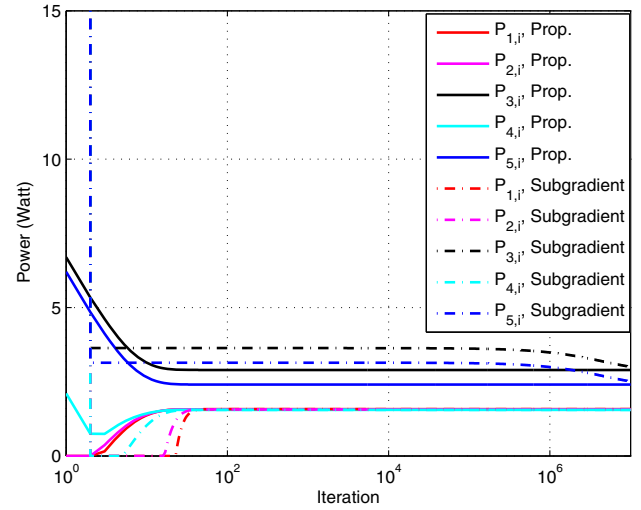


Fig. 4. Convergence of the proposed Algorithm 1 of the users powers connected to AP i and compared with the subgradient method, step size = 0.04, $P_{i,\max} = 10$ W.

selected carefully. It is known that in the subgradient method, the smaller the step size, the closer the optimal values we obtain as the number of iterations goes to infinity. Hence, as shown in Fig. 4, the proposed approach provides the same power values of the subgradient method (step size = 0.04) in less than 100 iterations, while the subgradient method achieves some of those values after 10^7 iterations. Figure 4 also shows that the subgradient method violates the maximum power constraints over a wide range of iterations, and it starts satisfying that constraint after a huge number of iterations (10 million iterations).

Figure 5 shows the convergence of Algorithm 2. It is clear that Algorithm 2 needs at most three iterations to converge. Furthermore, the number of iterations needed to converge does not depend on the number of users in the

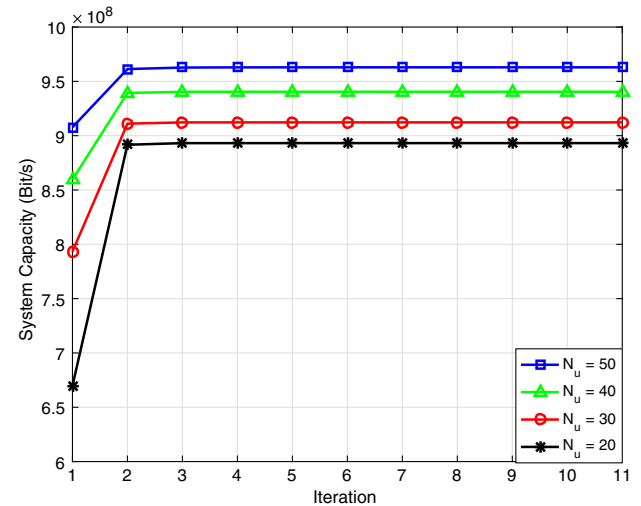


Fig. 5. System capacity versus number of iterations for Algorithm 2 with different numbers of system users.

system, which means that the number of users does not affect the convergence of Algorithm 2. This indicates that Algorithm 2 rapidly converges to the optimal solution. In addition, it can be seen from Fig. 5 that the value of the system capacity at iteration 1 is the result of allocating the power equally between users (Step 1 in Algorithm 2), and the value of the system capacity at iteration 2 is the result of solving the optimization problem, where the interference information is calculated from the power allocated in the first iteration, which is the equal power allocation, and so on. Therefore, Fig. 5 highlights the significant contribution of the proposed allocation power algorithm over allocating the power equally among users.

In Fig. 6, the relation between the system capacity and the number of reconnected users is shown with different numbers of total users. The number of transferred users means how many users transferred their connections from one AP to another. It can be seen that the more the users transfer, the better the system capacity until the system saturates. Also, it is obvious that more user transfers are needed for the system to reach the saturation point as the number of users increases, as expected. In other words, the rate of increasing the system capacity with 60 users is less than the rate of increasing the system capacity with 20 or 40 users.

The impact of the total number of users on the system fairness is studied in Fig. 7. It is clear from this figure that as the number of transferred users increases, the system fairness is more enhanced with the best results achieved at the minimum number of total users. Enhancing the system fairness and capacity together with each user transfer comes from the fact that the users with poor services travel from the overloaded APs to have better services from other less overloaded APs. This helps in decreasing the variance of the received data rate among users and provides a more efficient utilization of AP resources.

Figure 8 shows the relation between the number of users and the total system capacity for different values of α . Here,

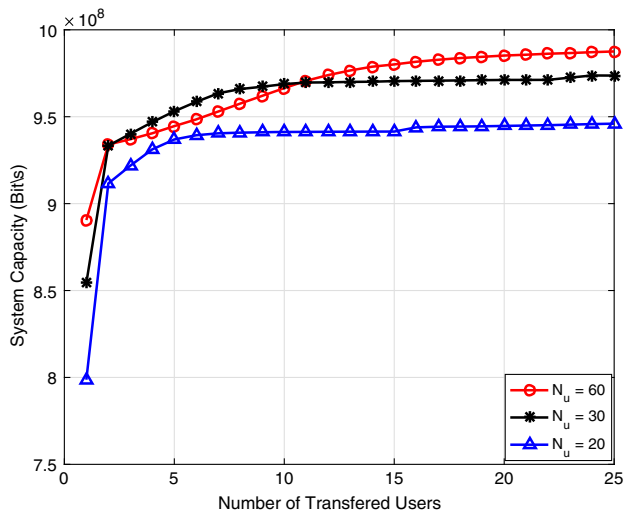


Fig. 6. System capacity versus the number of transferred users for different numbers of system users.

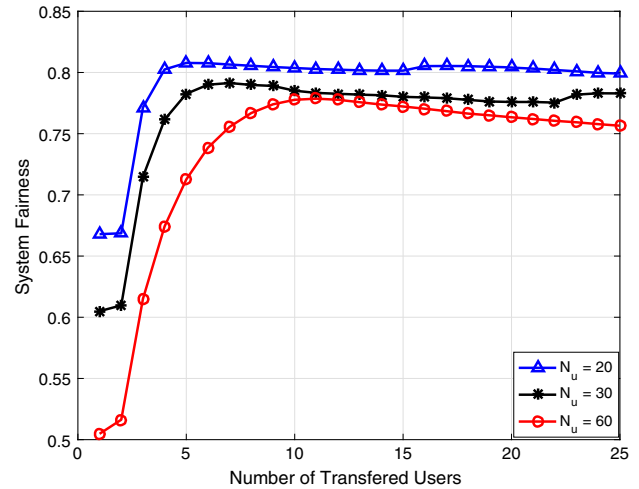


Fig. 7. System fairness versus number of transferred users for different numbers of system users.

$\alpha = 1$ means that no blockages happened, while $\alpha = 0.85$ means that the probability of having an object between APs and users is 0.15. It is shown that the difference between the performance of Algorithm 2 and the suboptimal solution (averaging the interference) is negligible. We also investigate the effect of the number of users on the system fairness in Fig. 9. The approach that considers the approximate interference information performs slightly better than the approach that considers the exact interference in terms of the system fairness. This is due to the fact that the former depends on calculating the interference matrix based on the assumption that the power at the AP is distributed fairly among users, which leads to a more fair power allocation among users than the latter. As can be seen from Figs. 8 and 9, as the probability of the LoS availability decreases, the system capacity increases while the fairness decreases. This is because increasing the blockage rate in the hybrid VLC/RF networks can enhance the

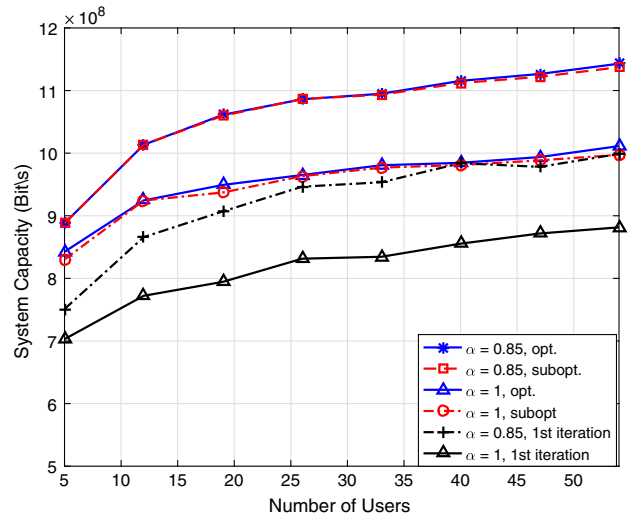


Fig. 8. Comparison between Algorithm 2 and the suboptimal approach in terms of system capacity for different values of α .

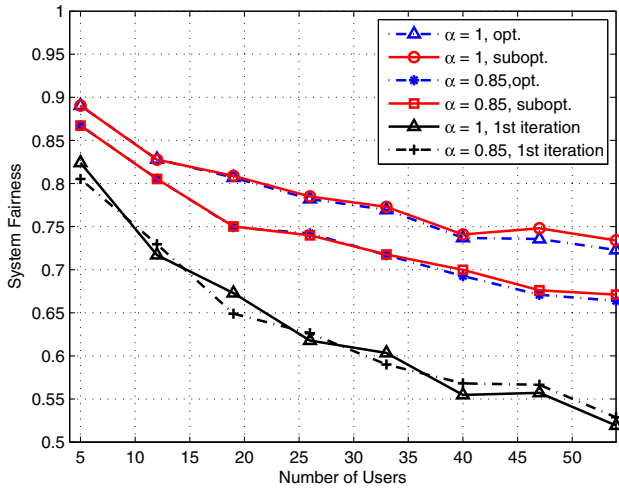


Fig. 9. Comparison between Algorithm 2 and the suboptimal approach in terms of system fairness for different values of α .

system capacity rather than compromise it. In other words, the blockages are more likely to prevent the users from receiving the optical interference than blocking the intended signals. On the other hand, blocking some users from some VLC APs increases the variance of the received data rates among users, which leads to a decrease in system fairness. The figures also show how the proposed joint LB and PA algorithms significantly improve the system capacity and fairness compared to the case where one iteration is implemented only (each user is assigned to its closest AP and each AP implements the PA only once).

V. CONCLUSION

In this paper, joint LB and PA algorithms for hybrid RF/VLC networks were proposed. An iterative algorithm was proposed to maximize the total system capacity and enhance the system fairness. Two approaches were implemented: 1) an approach that is based on the exact interference information for all users, which provided better performance, 2) and a suboptimal approach that is based on the approximate interference information, which had less complexity compared to the first approach. In the PA subproblem, we derived a new efficient algorithm that finds the optimal dual variables after formulating them in terms of each other. This new algorithm provided faster convergence and better performance than the traditional subgradient method. As future work, we recommend implementing the joint power and bandwidth allocation with load balancing in mobile users.

ACKNOWLEDGMENT

This work was funded by the Deanship of Scientific Research at the King Fahd University of Petroleum & Minerals (KFUPM) through grant number KAUST004. The work was also supported by the National Plan for Science, Technology, and Innovation (Maarifah)—King Abdulaziz City for Science and Technology—through the

Science and Technology Unit at KFUPM—the Kingdom of Saudi Arabia, under grant number 15-ELE4157-04. The authors would also like to acknowledge the KFUPM-KAUST research initiative that resulted from this research work. This work is an extended version of our paper, which has been recently accepted and presented at the IEEE GLOBECOM Conference 2017 [20].

REFERENCES

- [1] C.-L. Tsai and Z.-F. Xu, "Line-of-sight visible light communications with InGaN-based resonant cavity LEDs," *IEEE Photon. Technol. Lett.*, vol. 25, no. 18, pp. 1793–1796, 2013.
- [2] Y. Wang, D. A. Basnayaka, and H. Haas, "Dynamic load balancing for hybrid Li-Fi and RF indoor networks," in *IEEE Int. Conf. Communication Workshop (ICCW)*, London, UK, 2015, pp. 1422–1427.
- [3] M. Kashef, M. Abdallah, K. Qaraqe, H. Haas, and M. Uysal, "On the benefits of cooperation via power control in OFDM-based visible light communication systems," in *IEEE Personal, Indoor, and Mobile Radio Communication (PIMRC) Conf.*, Washington, DC, Sept. 2014, pp. 856–860.
- [4] M. B. Rahaim, A. M. Vegni, and T. D. Little, "A hybrid radio frequency and broadcast visible light communication system," in *IEEE Global Communication Workshops*, IEEE, 2011, pp. 792–796.
- [5] D. A. Basnayaka and H. Haas, "Hybrid RF and VLC systems: improving user data rate performance of VLC systems," in *IEEE Vehicular Technology Conf. (VTC Spring)*, Glasgow, UK, May 2015, pp. 1–5.
- [6] Y. Wang and H. Haas, "Dynamic load balancing with hand-over in hybrid Li-Fi and Wi-Fi networks," *J. Lightwave Technol.*, vol. 33, no. 22, pp. 4671–4682, Nov. 2015.
- [7] X. Wu, M. Safari, and H. Haas, "Bidirectional allocation game in visible light communications," in *IEEE Vehicular Technology Conf. (VTC Spring)*, Montreal, Canada, 2016, pp. 1–5.
- [8] M. Kashef, M. Ismail, M. Abdallah, K. A. Qaraqe, and E. Serpedin, "Energy efficient resource allocation for mixed RF/VLC heterogeneous wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 34, no. 4, pp. 883–893, 2016.
- [9] X. Li, R. Zhang, and L. Hanzo, "Cooperative load balancing in hybrid visible light communications and WiFi," *IEEE Trans. Commun.*, vol. 63, no. 4, pp. 1319–1329, 2015.
- [10] R. Zhang, H. Claussen, H. Haas, and L. Hanzo, "Energy efficient visible light communications relying on amorphous cells," *IEEE J. Sel. Areas Commun.*, vol. 34, no. 4, pp. 894–906, 2016.
- [11] X. Li, Y. Huo, R. Zhang, and L. Hanzo, "User-centric visible light communications for energy-efficient scalable video streaming," *IEEE Trans. Green Commun. Netw.*, vol. 1, no. 1, pp. 59–73, 2017.
- [12] D. W. K. Ng, E. S. Lo, and R. Schober, "Energy-efficient resource allocation in multi-cell OFDMA systems with limited backhaul capacity," *IEEE Trans. Wireless Commun.*, vol. 11, no. 10, pp. 3618–3631, 2012.
- [13] H. Wang, A. Liu, and X. Pan, "Optimization of joint power and bandwidth allocation in multi-spot-beam satellite communication systems," *Math. Prob. Eng.*, vol. 2014, pp. 683604, 2014.
- [14] J. Tang, D. K. C. So, E. Alsusa, K. A. Hamdi, A. Shojaeifard, and K.-K. Wong, "Energy-efficient heterogeneous cellular

- networks with spectrum underlay and overlay access," *IEEE Trans. Veh. Technol.*, vol. 67, pp. 2439–2453, 2017.
- [15] W.-C. Pao and Y.-F. Chen, "Adaptive gradient-based methods for adaptive power allocation in ofdm-based cognitive radio networks," *IEEE Trans. Veh. Technol.*, vol. 63, no. 2, pp. 836–848, 2014.
- [16] S. Xiao, X. Zhou, Y. Yuan-Wu, G. Y. Li, and W. Guo, "Robust resource allocation in full-duplex-enabled OFDMA femtocell networks," *IEEE Trans. Wireless Commun.*, vol. 16, no. 10, pp. 6382–6394, 2017.
- [17] J. M. Kahn and J. R. Barry, "Wireless infrared communications," *Proc. IEEE*, vol. 85, no. 2, pp. 265–298, 1997.
- [18] W. Guo, S. Wang, and X. Chu, "Capacity expression and power allocation for arbitrary modulation and coding rates," in *IEEE Wireless Communication and Networks Conf. (WCNC)*, Shanghai, China, 2013, pp. 3294–3299.
- [19] S. Boyd and L. Vandenberghe, *Convex Optimization* (Cambridge University, 2004).
- [20] M. Obeed, A. M. Salhab, S. A. Zummo, and M. S. Alouini, "Joint load balancing and power allocation for hybrid VLC/RF networks," in *IEEE Global Communication Conf. (GLOBECOM)*, Dec. 2017, pp. 1–6.



Mohanad Obeed (S'17) received a B.E. in Computer and Communication Engineering from Taiz University, Taiz, Yemen, in 2008, and a M.Sc. in Electrical Engineering from King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 2016. He is currently pursuing a Ph.D. at KFUPM, Dhahran, Saudi Arabia. His research interests include visible light communications, cooperative communication, resource allocation,

convex optimization, physical layer security, and energy harvesting.



Anas M. Salhab (S'11-M'14-SM'17) received a B.Sc. in Electrical Engineering from Palestine Polytechnic University, Hebron, Palestine, in 2004, a M.Sc. in Electrical Engineering from Jordan University of Science and Technology, Irbid, Jordan, in 2007, and a Ph.D. from King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 2013. From 2013 to 2014, he was a Postdoctoral Fellow with the Electrical

Engineering Department, KFUPM. He is currently an Assistant Professor and the Assistant Director of the Science and Technology Unit with the Deanship of Scientific Research, KFUPM. His research interests span special topics in modeling and performance analysis of wireless communication systems, including cooperative relay networks, cognitive radio relay networks, free-space optical networks, visible light communications, and co-channel interference. He was selected as an Exemplary Reviewer by the IEEE Wireless Communications Letters for his reviewing service in 2014. Recently, he received the KFUPM Best Research Project Award as a Co-investigator among the projects in 2013/2014 and 2014/2015.



Salam A. Zummo (M'00-SM'08) received B.Sc. and M.Sc. degrees in Electrical Engineering from the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 1998 and 1999, respectively, and a Ph.D. from the University of Michigan, Ann Arbor, USA, in 2003. He is currently a Professor with the Electrical Engineering Department, KFUPM. Prof. Zummo has six issued U.S. patents and has authored over 100 papers in reputable journals and conference proceedings. His research interests are in the area of wireless communications, including cooperative diversity, cognitive radio, multiuser diversity, scheduling, MIMO systems, error control coding, multihop networks, and interference modeling and analysis in wireless systems. He received the Saudi Ambassador Award for early Ph.D. completion in 2003, and the British Council/BAE Research Fellowship Awards in 2004 and 2006. He also received the KFUPM Excellence in Research Award from 2011 to 2012.



Mohamed-Slim Alouini (S'94, M'98, SM'03, F'09) was born in Tunis, Tunisia. He received a Ph.D. in Electrical Engineering from the California Institute of Technology (Caltech), Pasadena, California, USA, in 1998. He served as a faculty member at the University of Minnesota, Minneapolis, Minnesota, USA, then at the Texas A&M University at Qatar, Education City, Doha, Qatar before joining King Abdullah University of Science and Technology (KAUST), Thuwal, Makkah Province, Saudi Arabia as a Professor of Electrical Engineering in 2009. His current research interests include modeling, design, and performance analysis of wireless communication systems.